Heisenberg wasn't particularly concerned with solving the measurement problem. He was more concerned with getting another offer for a tenured professorship. He was worried that Schrödinger's accomplishments had eclipsed his own and that he had made a mistake in returning to Copenhagen rather than accepting the permanent professional safety that had been offered by Leipzig. Hungry for another major insight to improve his chances on the job market—and to one-up Schrödinger—Heisenberg turned his attention to measurement, but not the measurement problem. Instead, he focused on something less difficult and more likely to yield results: the limitations on what we can learn about quantum objects. Combining Born's new idea with some of Einstein's suggestions from their meeting in Berlin, Heisenberg uncovered a pithy new truth that, he thought, put the lie to Schrödinger's idea of an orderly quantum world.

Heisenberg started thinking about what would happen if you tried to measure the position of a single particle, like an electron, to very high precision. He realized that you could do this the same way you'd look for a lost wallet in a dark field: shine a flashlight around until you've found what you're looking for. An ordinary flashlight wouldn't work for an electron, though—the wavelength of visible light is far too large for that. But Heisenberg knew you could find an electron using higher-energy light, with a shorter wavelength: gamma rays. Shine a gamma-ray flashlight around the room, and you'll find your electron. But gamma rays pack a punch—bounce one gamma-ray photon off an electron, and the electron will go careening off in some random direction. So you'll know where the electron was, but you won't know how fast it's going or where it's heading now.

Heisenberg wondered if this kind of trade-off between measuring an object's position and its momentum was unavoidable, or if it was just an artifact of his thought experiment. To his delight, he discovered that these limits on measurement were fundamental: buried in the mathematics of Schrödinger's wave mechanics, Heisenberg found a precise formulation of how much information you have to give up about an object's momentum in order to learn more about its position, and vice versa. You could know a lot about where an object was or a lot about how it was moving—but you couldn't know both at the same time.

At Bohr's urging, Heisenberg used the term "uncertainty principle" to